Homework 6

1. **ARRANGEMENT with at least two consecutive identical pairs.**

   The word `ARRANGEMENT` has 11 letters with multiplicities

   $$A^2,\ R^2,\ N^2,\ E^2,\ G,\ M,\ T.$$

   For simplicity, we wish to utilize the fancy symbols to denote adjacent characters defined as follows:

   $$\mathcal{A} = \{AA \text{ adjacent}\},\ \mathcal{R} = \{RR \text{ adjacent}\},\ \mathcal{N} = \{NN \text{ adjacent}\},\ \mathcal{E} = \{EE \text{ adjacent}\}.$$

   We wish to find the number of permutations in which at least two of these four events occur.

   Trivially, we recognize if we force a chosen set of s pairs to be adjacent, we treat each such pair as a single block. This reduces the number of objects from 11 to $11 - s$. We have the following cases:

   - If $s = 1$ (say AA is a block), we permute

     $$(AA),\ R, R,\ N, N,\ E, E,\ G, M, T$$

     which is 10 objects with three remaining duplicated letter-types, so

     $$|\mathcal{A}| = \frac{10!}{2!2!2!} = \frac{10!}{8} = 453600.$$

     Hence $a_1 = \binom{4}{1} \cdot |\mathcal{A}| = 4 \cdot 453600 = 1814400$.
   - If $s = 2$, we have 9 objects and two duplicated types remaining, so for any fixed pair of events,

     $$|\mathcal{A} \cap \mathcal{R}| = \frac{9!}{2!2!} = \frac{9!}{4} = 90720.$$

     Thus $a_2 = \binom{4}{2} \cdot 90720 = 6 \cdot 90720 = 544320$.
   - If $s = 3$, we have 8 objects and one duplicated type remaining:

     $$|\mathcal{A} \cap \mathcal{R} \cap \mathcal{N}| = \frac{8!}{2!} = 20160,$$

     so $a_3 = \binom{4}{3} \cdot 20160 = 4 \cdot 20160 = 80640$.
   - If $s = 4$, we have 7 distinct blocks/letters:

     $$|\mathcal{A} \cap \mathcal{R} \cap \mathcal{N} \cap \mathcal{E}| = 7! = 5040,$$

     so $a_4 = 5040$.

   For greater simplicity, we wish to define a N_s which is the number of arrangements with exactly s of the four adjacency-events occurring. Then, we have:

   $$N_4 = a_4 = 5040,$$

and since each arrangement counted in N_4 lies in all $\binom{4}{3} = 4$ triple-intersections,

$$N_3 = a_3 - 4N_4 = 80640 - 4 \cdot 5040 = 60480.$$

Trivially, we find a_2 counts each arrangement in N_2 exactly once, each arrangement in N_3 in $\binom{3}{2} = 3$ ways, and each arrangement in N_4 in $\binom{4}{2} = 6$ ways, so

$$a_2 = N_2 + 3N_3 + 6N_4 \quad \Longrightarrow \quad N_2 = a_2 - 3N_3 - 6N_4.$$

Therefore, we have:

$$N_2 = 544320 - 3(60480) - 6(5040) = 544320 - 181440 - 30240 = 332640.$$

And so, after extensive computation that likely could have remained in its original form, the number with at least two adjacent identical pairs is

$$N_2 + N_3 + N_4 = 332640 + 60480 + 5040 = \boxed{398160}.$$

2. **Combinatorial proof that $n! = \binom{n}{0}(!0) + \binom{n}{1}(!1) + \cdots + \binom{n}{n}(!n)$.**

   We wish to define $[n] = \{1, 2, \ldots, n\}$. We count all permutations of $[n]$ in two ways.

   (1) Trivially, this is a count. There are $n!$ permutations of $[n]$.

   (2) Count by the set of moved elements. Given a permutation π, let $S = \{i \in [n] : \pi(i) \neq i\}$ be the set of elements *not fixed* by π. Suppose $|S| = k$. Then:

   - Choose which k elements are moved: $\binom{n}{k}$ choices.
   - On those k chosen elements, π must have no fixed points (otherwise that element would not be in S). The number of permutations of k elements with no fixed points is the derangement number $!k$.
   - The remaining $n - k$ elements are fixed automatically.

   Therefore, the number of permutations with exactly k moved elements is $\binom{n}{k}(!k)$. Summing over $k = 0, 1, \ldots, n$ counts all permutations, so

   $$n! = \sum_{k=0}^{n} \binom{n}{k}(!k) = \binom{n}{0}(!0) + \binom{n}{1}(!1) + \cdots + \binom{n}{n}(!n).$$

   QED.

3. **Returning two quizzes and an exam to 25 students so nobody gets any of their own papers.**

   Each assignment is distributed separately. For one fixed assignment (say Quiz 1), returning papers corresponds to a permutation of the 25 papers among the 25 students. The condition that *no student gets their own paper* means the permutation has no fixed points, i.e. it is a derangement. Therefore, the number of ways for one assignment is $!25$.

   Since the two quizzes and the exam are distributed independently, by the product rule the total number of ways is

   $$\boxed{(!25)^3}.$$

4. **Rook polynomials.** From lecture notes, we know for a board C, the rook polynomial is

$$r(C, x) = \sum_{k \geq 0} r_k x^k,$$

where r_k is the number of ways to place k nonattacking rooks on C.

(a) **The 3×3 board.**
To place k nonattacking rooks on a 3×3 board: choose k rows ($\binom{3}{k}$ ways), choose k columns ($\binom{3}{k}$ ways), then match the chosen rows to chosen columns (a bijection), which is $k!$ ways. Therefore, we find

$$r_k = \binom{3}{k}^2 k! \qquad (k = 0, 1, 2, 3).$$

Which leads to

$$r_0 = 1, \quad r_1 = \binom{3}{1}^2 1! = 9, \quad r_2 = \binom{3}{2}^2 2! = 3^2 \cdot 2 = 18, \quad r_3 = \binom{3}{3}^2 3! = 6.$$

And so we arrive at

$$\boxed{r(x) = 1 + 9x + 18x^2 + 6x^3.}$$

(b) **The standard 8×8 chessboard.**
The same reasoning gives for $k = 0, 1, \ldots, 8$:

$$r_k = \binom{8}{k}^2 k!.$$

Therefore

$$\boxed{r(C, x) = \sum_{k=0}^{8} \binom{8}{k}^2 k!\, x^k.}$$

If we want to use desmos and expand this, it's:

$$\boxed{r(C, x) = 1 + 64x + 1568x^2 + 18816x^3 + 117600x^4 + 376320x^5 + 564480x^6 + 322560x^7 + 40320x^8.}$$